## Step 1: Open Workbench

- Open **MySQL Workbench**
- Click your local connection (usually `Local instance MySQL`)

## 🔹 Step 2: Open Query Tab

Click:
👉 **SQL + icon (New Query Tab)**

## 🔹 Step 3: Paste your code

```
CREATE DATABASE librarydb;
USE librarydb;

CREATE TABLE books (
  id INT PRIMARY KEY AUTO_INCREMENT,
  title VARCHAR(100),
  author VARCHAR(100),
  price DOUBLE
);

INSERT INTO books (title, author, price) VALUES
('Java Programming', 'James Gosling', 500),
('DBMS Concepts', 'Korth', 450),
('Web Tech', 'Deitel', 400);
```

## 🔹 Step 4: Run

Click:
👉 ⚡ **Execute button**

## 🔹 Step 5: Verify

On left side:

- Refresh schemas
- Open:

librarydb → tables → books

- Right-click → **Select Rows**

👉 You should see your data

# Step 2.1: Download Connector

Go to:
👉 https://dev.mysql.com/downloads/connector/j/

Download:
👉 **Platform Independent (ZIP)**

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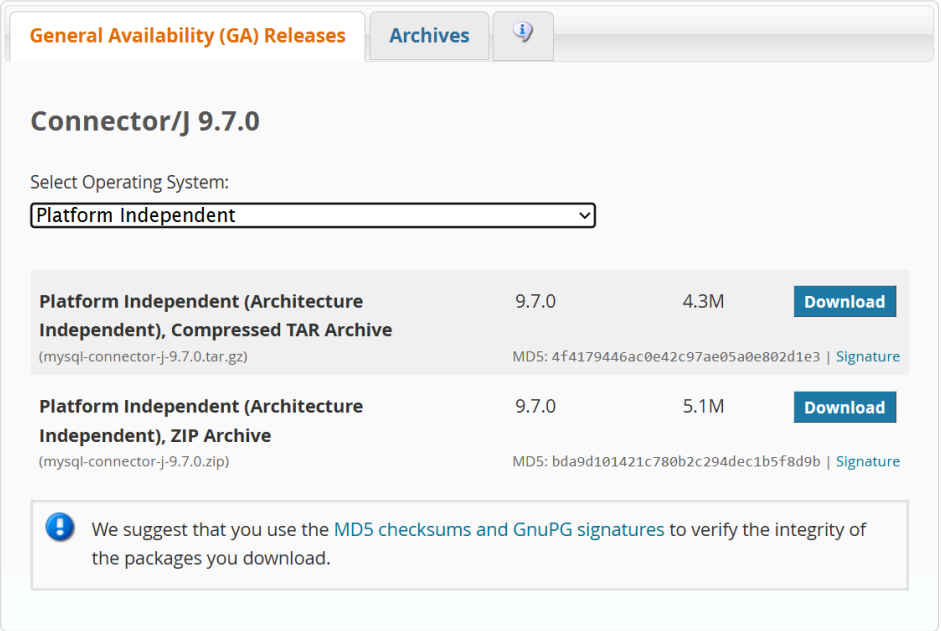


# Step 2.2: Extract

- Extract ZIP
- Inside you'll find:

mysql-connector-j-8.x.x.jar

# Step 2.3: Add to Eclipse Project

### Do this exactly:

1. Right-click:
   👉 `DemoApp`

2. Click:
   👉 **Build Path → Configure Build Path**
3. Go to:
   👉 **Libraries tab**
4. Click: **class path u have to add**
   👉 **Add External JARs**
5. Select:
   👉 `mysql-connector-j-8.x.x.jar`
6. Click:
   👉 **Apply → OK**

### 🔹 Step 2.4: Verify (IMPORTANT)

After adding:

- Expand project
- You should see:

  Referenced Libraries → mysql-connector-j...

# STEP 3: Create Servlet

Now:

1. Right-click:
   👉 `src/main/java`

## Use “Other…” option

From your screenshot:

👉 Click:
**New → Other… (Ctrl + N)**

### 🔹 Step 2: Search

In the popup:

Type:

Servlet

## 🔹 Step 3: Select

Choose:
👉 **Web → Servlet**

Click **Next**

## 🔹 Step 4: Fill details

- Package: `com.demo`
- Class Name: `BookServlet`

Click **Finish**

**Copy this code in bookservlet.java:**

```
package com.demo;

import java.io.*;

import java.sql.*;

import jakarta.servlet.*;

import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.*;

@WebServlet("/books")

public class BookServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<h2>Book Details</h2>");

        out.println("<table border='1'>");

        out.println("<tr><th>ID</th><th>Title</th><th>Author</th><th>Price</th></tr>");
```

```
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/librarydb", "root", "password");
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM books");
            while (rs.next()) {
                out.println("<tr>");
                out.println("<td>" + rs.getInt("id") + "</td>");
                out.println("<td>" + rs.getString("title") + "</td>");
                out.println("<td>" + rs.getString("author") + "</td>");
                out.println("<td>" + rs.getDouble("price") + "</td>");
                out.println("</tr>");
            }
            con.close();
        } catch (Exception e) {
            out.println("<p>Error: " + e.getMessage() + "</p>");
        }
        out.println("</table>");
    }
}
```

Index.html:

```
<!DOCTYPE html>
<html>
<body>
```

```
<h1>Book System</h1>

<a href="books" target="frame">View Books</a>

<br><br>

<iframe name="frame" width="100%" height="400px"></iframe>

</body>

</html>
```

## Step 1: Add JAR to Deployment Assembly

1. Right-click:
   👉 `DemoApp`
2. Click:
   👉 Properties

## 🔹 Step 2: Go to

👉 Deployment Assembly

## 🔹 Step 3: Click

👉 Add…

## 🔹 Step 4: Select

👉 Java Build Path Entries

Click Next

## 🔹 Step 5:

✔ **Tick:**
👉 **mysql-connector-j-xxx.jar**

**Click Finish**

## 🔹 Step 6:

**Click:**
👉 **Apply → OK**

Right-click:
👉 **DemoApp**

Then:
👉 **Run on Server**

Select Tomcat → Finish

# Book System

View Books

## Book Details

| ID | Title | Author | Price |
|---|---|---|---|
| 1 | Java Programming | James Gosling | 500.0 |
| 2 | DBMS Concepts | Korth | 450.0 |
| 3 | Web Tech | Deitel | 400.0 |